

CY 200: Foundations and Applications of Chemistry

Course Code: CY 200

Course Name: Foundations and Applications of Chemistry

L-T-P-C: 3-0-0-3

Prerequisites: -

Students intended for: BS Chemical Sciences

Elective or Core: Core

Mutual Exclusion: -

Approval: 59th BoA

Course contents

- **The Evolution of Chemistry:** History of chemistry; Evolution of chemistry starting from the alchemist era; Modern chemistry; Significant discoveries in chemistry with Nobel Prizes; Chemistry for society; Opportunities in chemistry. [6 Lectures]
- **Chemistry in Nature:** Biomolecular Chemistry: Structure and function of carbohydrates, lipids, proteins, and nucleic acids; Enzyme catalysis and metabolic pathways; Carbon, nitrogen, and sulfur cycles; Chemical transformations in ecosystems; Natural products: Alkaloids, terpenes, and polyketides; Photosynthesis and Energy Cycles: Chemical principles of photosynthesis, Carbon fixation and the Calvin cycle; Energy flow in nature: ATP and biochemical energy carriers; Chemistry of Natural Materials: Chemistry of bones, shells, and coral reefs, Cellulose, chitin, and lignin, Chlorophyll, carotenoids, and melanin, etc. [12 Lectures]
- **Modern Applications of Chemistry:** Chemistry in Energy and Environment: Fuels and combustion, Renewable energy, Environmental chemistry; Chemistry in Materials Science: Polymers, Nanomaterials, Semiconductors, OLEDs; Chemistry in Biology and Medicine: Biochemical molecules, Proteins, DNA, and enzymes, Pharmaceuticals, Drug design; Diagnostic tools: Chemical sensors and imaging; Industrial and Everyday Applications: Catalysis in industrial processes, Chemistry of household product, Food chemistry. [12 Lectures]
- **Chemical Data Analysis with Software:** Types of chemical data: Spectroscopic, kinetic, thermodynamic, and structural; Overview of software tools for chemical data visualization and analysis with examples: ChemDraw, Origin, Excel, ChemCraft, MNova, Mercury, etc.; Statistical Methods in Chemical Data Analysis: Mean, median, standard deviation, and outliers, regression analysis, and curve fitting; Chemoinformatics and Databases: Molecular representations (SMILES, InChI), Databases (SciFinder, Reaxys, Google Scholar, PubChem, ChemSpider, and ChEMBL, etc.), Ethics of scientific practices. [8 Lectures]
- **Machine Learning in Chemistry:** Overview of machine learning concepts: Supervised, unsupervised, and reinforcement learning; The role of ML in modern chemical research; Examples of ML applications in chemistry. [4 Lectures]

References: (Books and articles)

1. Lowe D.B., **The Chemistry Book: From Gunpowder to Graphene, 250 Milestones in the History of Chemistry**, Sterling Publishing Company.



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2. Arthur Greenberg, **A Chemical History Tour: Picturing Chemistry from Alchemy to Modern Molecular Science**, Wiley-Interscience.
 3. David L. Nelson and Michael Cox, **Lehninger Principles of Biochemistry**, 8th Edition, Macmillan Learning.
 4. Fredus Nelson Peters, **Modern Chemistry, With Its Practical Applications**, Legare Street Press.
 5. D. Brynn Hibbert, and J. Justin Gooding, **Data Analysis for Chemistry: An Introductory Guide for Students and Laboratory Scientists**, Oxford University Press.
 6. Hugh M Cartwright, **Machine Learning In Chemistry: The Impact Of Artificial Intelligence**, Vol 17, Royal Society of Chemistry.